

Progress Test 2.C — Thermo-S1

2.C · Thermoelastic coefficients: principle and method
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Quick calculation (1 mark)

At $p = 1.0 \times 10^5 \text{ Pa}$, a gas has $\alpha = 3.3 \times 10^{-3} \text{ K}^{-1}$ and $\chi_T = 1.0 \times 10^{-5} \text{ Pa}^{-1}$. Give its piezothermic coefficient β .

[Empty grid for Exercise 1 answer]

Marking

①
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Exercise 2. Relation between the coefficients (5 marks)

State the cyclic relation between the partial derivatives of p , V and T . Use it to show that $\beta = \frac{\alpha}{p \chi_T}$.

[Empty grid for Exercise 2 answer]

Marking

①
 0 1 2
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 0 1 2
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 ③
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Exercise 3. Method in thermodynamics

(3 marks)

To obtain the equation of state from α and χ_T , one must:

- ☐ differentiate ☐ integrate ☐ multiply them ☐ average them

To obtain α from a known equation of state, one must:

- ☐ differentiate ☐ integrate ☐ invert it ☐ plot it

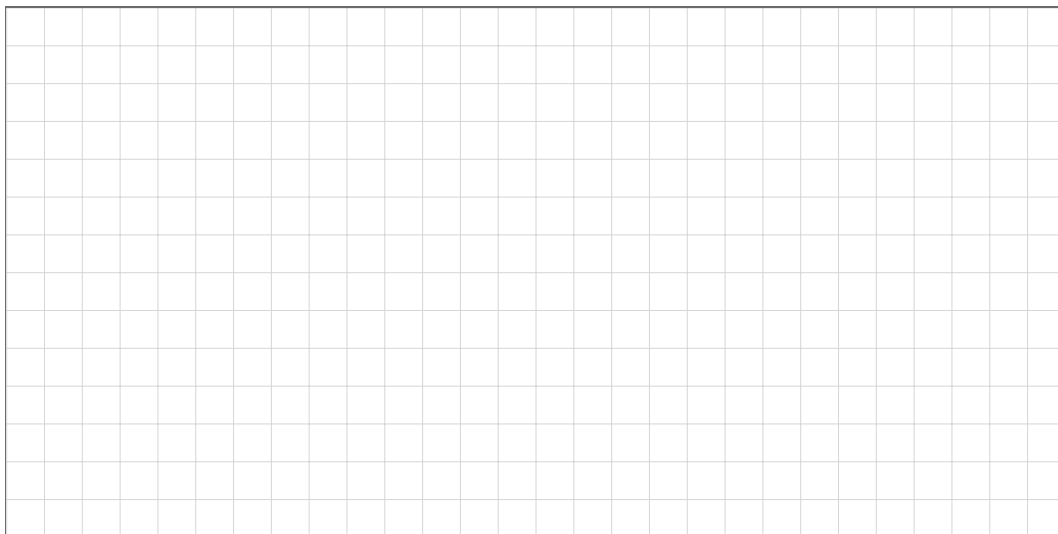
The three coefficients α , χ_T and β are:

- ☐ independent of each other ☐ linked by $\beta = \alpha/(p\chi_T)$
☐ always equal ☐ all dimensionless

Exercise 4. From an equation of state to the coefficients

(5 marks)

A gas obeys the equation of state $p(V - nb) = nRT$, where the co-volume b is a positive constant. Compute α , χ_T and β , then check that $\beta = \alpha/(p\chi_T)$. What do you find when $b \rightarrow 0$?

**Marking**

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— End of the test —